

Mai Thế Toàn

Backend Developer

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CAREER OBJECTIVE

Backend Developer focused on ASP.NET Core, distributed systems, and scalable API design. Experienced in building systems, real-time communication, and AI-powered applications using Redis, RabbitMQ, Docker, and Clean Architecture.

EDUCATION

Vietnam National University, Hanoi, VNU	2023 - 2026
Computer Science	
GPA: 3.5/4.0	

RESEARCH & PUBLICATIONS

Mai The Toan, Cho Do Xuan, Nguyen Manh Son.
"Enhancing Web Attack Detection Efficiency Based on Natural Language Processing Techniques."
Published in The Journal of Computer Science and Cybernetics, 2025.
DOI: 10.15625/1813-9663/23407
Role:
- Conducted NLP model research
- Built web attack classification datasets
- Evaluated detection performance
Researched and trained LDWL-based NLP models on a dataset of 31,067 web requests, achieving 99.99% Accuracy and F1-Score in detecting common web attacks, including SQL Injection and XSS. Designed the solution for future integration as a security middleware for Web APIs.

PROJECTS

Calendar Bot	01/2026 - 05/2026
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AI SMART APPOINTMENT CHATBOT

- Architected and implemented 20+ APIs for appointment scheduling and conflict detection. Optimized response latency from 500ms to under 50ms using Redis caching and leveraged RabbitMQ for asynchronous notification and AI processing, improving scalability and high-availability.
- Tech Stack:** ASP.NET Core, SQLServer, Redis, RabbitMQ, Docker, OpenAI API, Clean Architecture, SignalR (Websocket), Docker compose.
- Github:** <https://github.com/Toandz1125/CalendarBot>

Sort Schedule	03/2025 - 01/2026
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UNIVERSITY SCHEDULING MANAGEMENT SYSTEM

- Engineered a university-wide scheduling system using Tabu Search, Graph Coloring, and First-Fit algorithms, solving 178+ conflict-free schedules while satisfying classroom, lecturer, and time-slot constraints. Reduced timetable planning time from weeks to minutes.
- Tech Stack:** ASP.NET Core, EF Core, SQL Server, JWT, Clean Architecture.
- Github:** <https://github.com/Toandz1125/SortSchedule>

Pharmacy Management	06/2024 - 03/2025
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PHARMACY MANAGEMENT SYSTEM

- Implemented inventory, sales, supplier, and customer management modules with support for 1,000+ product records and transaction processing workflows.
- Tech Stack:** C#, WinForms, SQL Server, Entity Framework.
- Github:** <https://github.com/Toandz1125/LTNC/tree/main/PharmacyManagement>

Crypto Currency Tickers	09/2023 - 06/2024
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CRYPTO CURRENCY TICKERS

- Built an ESP32-based cryptocurrency ticker displaying real-time prices via the CoinGecko API with Wi-Fi connectivity and JSON data processing. The project was selected for exhibition during the university anniversary celebration.
- Tech Stack:** ESP32, C++, Wi-Fi Communication, REST API, CoinGecko API, JSON parsing, Embedded Systems.
- Github:** <https://github.com/Toandz1125/Crypto-price-tickers>

SKILLS

Programming Languages C#, SQL, Javascript, Python, C++

Backend Development	ASP.NET Core, ASP.NET Web API, RESTful API Design, WinForms, Middleware, Dependency Injection, Transactions
Database	SQL Server, PostgreSQL, Entity Framework Core, LINQ, Database Design, Stored Procedure, Query Optimization
Authentication & Security	JWT Authentication, ASP.NET Identity, Role-Based Authorization
Architecture & Design	Clean Architecture, Repository Pattern, SOLID Principles, MVC Pattern, DDD (Domain-Driven Design)
DevOps	Docker, GitHub Actions, CI/CD
Monitoring	Prometheus, Grafana
Tools	Git, GitHub, Docker, Postman, Redis cache, RabbitMQ, SignalR, Visual Studio, AI coding assistant, Windows Terminal, PythonAnywhere, Firebase, Cloudflare, Google Cloud, Arduino, Canva

HONORS & AWARDS

Outstanding Student Award	2023
Awarded Academic Excellence Scholarship in Every Semester Throughout Undergraduate Studies	

CERTIFICATES

IELTS 6.0	2022
JLPT N4	2024

HOBBIES

Researching, Running, Music, Guitar, Sports